Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Diode Clippers : Parallel using Ideal diode

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Diode Clippers : Parallel using Ideal diode

1 Diode Clippers : Parallel using Ideal diode

Study focus: Diode Clippers : Parallel using Ideal diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000082 · EV0000083

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Professor Ananda, Department of Electronics and Communication Engineering.

Key details

- Today, we're going to see the parallel clipper, but the last class we have seen the series clipper.
- The series clipper are those which is used to clip off the waveform that is not required and without disturbing the other part of the waveform.
- The analysis of the parallel configuration is very similar to that of the series configuration.
- The other end of the resistor is connected to the diode.
- When the diode is on, we represent it by a short circuit.
- When the diode is off, represented as an open circuit.
- When it's represented as an open circuit, the voltage across the diode is same as the input applied.
- The same thing applies to any input in the place of a square if you use the triangle as shown.
- Determine the Vout for the network of the figure 16.
- DC voltage, V is equal to 4 volts, with the positive side connected to the positive side of the diode and negative side connected to the ground.
- And the diode positive side is connected to the resistor, where the voltage applied.
- VI is applied to the one end of the resistor, other end is connected to the supply voltage.
- DC voltage, and we see that the negative is in series with the positive.
- That means the applied voltage here will get minus with the V.
- That means to say...
- So we'll try to understand that with the next step.
- Then, in other words, when the diode is on, the output will be 4 volts.

- This we need to prove now.
- Then when the diode is in open circuit, that is the off state, the current through the series network will be 0, and the voltage drop across the resistor will be 0.
- That will be Vout is equal to VI.
- During the positive half cycle, transition from 0 to T/2, diode is on.
- How do we represent?
- 4 volts there.
- And the diode is 0, and transition 0 instead of transition.
- That is why VI greater than 4 volts, Vout=VI.
- That means to say, if VI is greater than 4 volts, then VA greater than 4 volts, diode is off.
- How do you represent that?

Example or application from the lecture

- Let us have a refresh of that.
- Now let us go ahead with the parallel clipper, first without bias, then with bias.
- So let us go with the clipper with a bias, with an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 05:00–09:59

Greater than 4, less than 4, 4 volts, greater than 4.

Key details

- VK is equal to 0.7.
- During the transition voltage, using the ID is equal to 0 or VD is equal to 0.7 volts and draw current in the network as shown in the figure 19.
- In the previous case, it was 0, VD was 0.
- The total voltage applied is, we'll add or subtract, so 3.3.
- It's subtract here because I have minus plus or plus minus.
- It's 3.3.
- In the previous case, it was 4 volts there.
- Whereas now, the total voltage, sum of the voltage here is 4 minus 0.7 is 3.3 we get.
- Then during the negative half cycle what we notice is, same as an input.
- The input voltage is 16 here, and hence we follow up to 16 the same waveform, and T/2 to T, the same 3.3 will be continued.
- The summary of the series clipper and the parallel clipper.
- In a positive clipper, the positive part is missing here.
- The circuit is this.
- Hence we call it as a negative clipper.
- Then, coming to the bias.
- The positive was missing in the previous case.
- Now positive is there, but it got clipped off because of the voltage available here.
- Presence of the voltage over here.

- This is because of the transition happened because of the clipper.
- Because of the voltage applied there.
- Then come to the simple parallel clipper with ideal diodes.
- The positive part is missing, positive clipper.
- Negative part is missing, negative clipper.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Diode Clippers : Parallel using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9977; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode Clippers : Parallel using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...
SRC00040	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...
SRC00041	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...

C Revision and retrieval practice